

Fluid Independence: Powered Wheelchair Users' Perspectives on Autonomy, Care, and Assistive Technology

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Abstract

Advancements in assistive technologies promise greater independence for wheelchair users; however, lived realities reveal a complex interplay of empowerment, dependency, and unmet needs. Accessibility discourse still leans heavily on clinical or technical perspectives, underweighting the lived expertise of powered-wheelchair users. From 55 in-depth interviews, we demonstrate that independence is fluid and continually negotiated across various contexts, including caregiving dynamics, privacy trade-offs, and structural barriers. Technologies that reduce frequent “ask-points” (e.g., retrieving objects, opening doors) deliver disproportionate autonomy gains, while caregiver shortages and insurance denials persist. Dependency was experienced as intimacy, gratitude, frustration, and loss of dignity, underscoring the emotional stakes of technology adoption. Participants also heavily relied on peer networks and custom solutions to navigate technology choices. Based on our insights, we introduce an ecosystem-oriented framework and user-led directory (*Assistive-Tech Commons*), positioning wheelchair users as co-designers. Centering emotional well-being, household ecologies, and community knowledge enables more sustainable, dignity-preserving assistive technologies.

CCS Concepts

• **Human-centered computing** → *Accessibility*.

Keywords

Powered-Wheelchair Users' Experiences, Assistive Technology, Independence, Accessibility, Participatory Design

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1 Introduction

A wheelchair represents more than mobility; it embodies autonomy, dignity, and full societal participation [80]. Over 70 million individuals worldwide depend on wheelchairs, and powered wheelchairs have transformed possibilities for users to engage meaningfully in work, education, and social life [2]. Rapid advancements in assistive technology, from sophisticated seating solutions to advanced environmental interfaces, promise unprecedented levels of personal independence. However, the landscape of assistive technologies today presents a paradox. While technical capabilities have significantly advanced, their real-world impact remains constrained by deeper structural problems, including prohibitive costs, fragmented healthcare and support systems, insufficient participatory training, and inaccessible physical environments [65]. These barriers are understood not as isolated inconveniences but as properties of the health and social care systems that wheelchair users must continually navigate, shaping when and how assistive technologies can be used meaningfully [72]. Moreover, the dependency on caregivers, whether family members, paid professionals, or institutional personnel, introduces critical and overlooked layers of complexity, influencing users' practical autonomy and emotional and social well-being. The distinction between thriving and simply coping often relies less on the inherent capabilities of a device and more on how it is integrated into users' daily routines, social relationships, and broader support structures [76, 77]. The lives of powered-wheelchair users are embedded in what systemic service design scholars describe as multi-actor service ecosystems [73], where policies, funding arrangements, and care infrastructures co-produce or constrain the possibilities for independence.

The field of HCI has increasingly recognized the limitations of purely technical solutions and has shifted toward a more holistic, participatory and inclusive approach to assistive technologies [20, 40, 68]. However, much of this work still treats devices and interfaces as the primary unit of analysis, with less attention to how health systems, policy infrastructures, and caregiving arrangements jointly shape what technologies can do in practice. Despite growing advocacy for inclusive design, research and development



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efforts frequently center on the perspectives of clinicians and technologists, rather than directly engaging wheelchair users as equal contributors to innovation processes. This oversight is significant, as insufficient end-user involvement contributes to device abandonment, diminished effectiveness, unmet daily needs, and reduced quality of life [57].

Furthermore, assistive technology's emotional and psychological dimensions, such as frustration, stigma, empowerment, or gratitude, remain underexplored, even though they profoundly influence adoption and sustained use [11, 68]. Nevertheless, questions remain about how powered-wheelchair users themselves experience the everyday tensions between autonomy and caregiving, and how they adopt and adapt assistive technologies within the social, cultural, and infrastructural contexts that shape their lives. These gaps call for a systemic perspective that treats autonomy as emerging from the coupled dynamics of user agency, caregiving ecologies, technology affordances, and structural contexts such as insurance, policy, and environmental accessibility, rather than from any single device or interaction. To address these gaps, we ask one overarching research question:

RQ: How do powered-wheelchair users negotiate independence as they navigate autonomy, caregiving, and assistive technologies within structurally constrained ecosystems?

We explore this overarching question through three focused sub-questions:

- **RQ1:** In what ways do powered-wheelchair users experience and define independence and autonomy in the context of daily caregiving arrangements?
- **RQ2:** How do powered-wheelchair users adopt, adapt, and integrate assistive technologies into their daily routines, and what barriers and enablers within their care ecologies influence these processes?
- **RQ3:** How do broader social, policy, and infrastructural contexts (e.g., insurance, infrastructure) mediate both caregiving arrangements and the long-term use of assistive technologies, and what implications does this have for designing more systemic, ecosystem-oriented interventions?

We approached these research questions through 55 semi-structured interviews with powered-wheelchair users across various ages, conditions, and caregiving arrangements. The interviews shed light on how autonomy is negotiated in everyday life, not only through the use of devices but also in relation to caregiving practices and social interactions. Our analysis illustrates often-overlooked dynamics, including the role of peer networks, the emotional weight of dependency, and persistent gaps in training and support. These findings highlight that the adoption and use of assistive technologies cannot be reduced to questions of device functionality, but must be understood within the broader emotional, relational, and institutional contexts that shape user empowerment.

Thus, we make the following contributions: (1) we offer an empirically grounded comprehensive understanding of the lived experiences of powered-wheelchair users, showing how emotional, relational, and systemic factors shape the use and meaning of assistive technologies; (2) we translate these insights into holistic and participatory design implications, foregrounding caregiving dynamics and emotional wellbeing as central to technology adoption and

long-term use; and (3) we propose an ecosystem-oriented, user-led framework that positions powered-wheelchair users as co-creators rather than recipients of technology, offering recommendations for HCI researchers, designers, and policymakers working toward more inclusive and empowering forms of assistive technology.

2 Related Work

To contextualize our study, we reviewed prior work on assistive technology models, the emotional dimensions of autonomy and care, distributed decision-making in technology adoption, interview-based studies with powered-wheelchair users, and participatory and disability justice approaches.

2.1 Foundational Models of Assistive Technology Use and Their Socio-Technical Limitations

The field of assistive technology (AT) has long relied on conceptual models to guide assessment and intervention, moving beyond medical or device-centric perspectives. Seminal frameworks such as the Human Activity Assistive Technology (HAAT) model, the Student–Environment–Tasks–Tools (SETT) framework, and ecological models from occupational therapy conceptualize AT use as an interaction among people, tasks, tools, and contexts [1, 14, 32, 36, 45, 50, 82]. By emphasizing the fit between user, activity, and environment, these approaches provided a foundation for later user-centered work in HCI.

However, these frameworks function primarily as static “snapshot” tools. They support point-in-time assessments but do not capture the longitudinal processes of procurement, funding, training, adaptation, and maintenance that shape lived experiences. This misalignment contributes to discontinuance rates of nearly 30%, where devices are abandoned not because of technical failure but because they fail to align with evolving routines, identity, or social environment [50, 55, 61]. A further critique is that socio-cultural and affective dimensions, though theoretically included in models like HAAT, are rarely applied in practice. Reviews note a “lack of studies” addressing these dimensions, reflecting a disciplinary inertia that prioritizes technical functionality over emotional and relational realities of AT use.

Comparisons with information systems research underscore this limitation. Models such as the Technology Acceptance Model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT) theorize adoption as a socially embedded process, integrating constructs like social influence, subjective norms, and moderating variables such as age and gender [22, 41, 79]. These models demonstrate that technologies succeed or fail partly because of how they are endorsed within social networks, a factor largely overlooked in AT research. Alternative frameworks offer ways forward. Sen's Capability Approach, extended into communication contexts through the Communication Capability Approach, reframes AT not as restoring lost function but as expanding fundamental freedoms and valued “beings and doings” [33, 69]. This lens positions users' choices, even discontinuing device use, as agentic and legitimate rather than failures [48], aligning with our focus on fluid, negotiated independence and underscoring the importance of

privileging user expertise and self-determination over compliance with prescribed technologies.

In parallel to these conceptual and socio-technical models, AT outcomes have also been studied through standardized quality-of-life (QoL) instruments. For example, the World Health Organization Quality of Life (WHOQOL) assessment provides a widely used multidimensional measure of well-being that has been applied to examine how mobility, environmental accessibility, and walkability relate to QoL for people who use wheelchairs and other mobility aids [37]. While such scales are valuable for capturing aggregate outcomes at the level of domains (e.g., physical health, social relationships, environment), they typically offer a snapshot view of QoL and do not reveal the fine-grained, moment-to-moment negotiations around caregiving, privacy, and “ask-point” that our study foregrounds. Our work is therefore complementary: we unpack the everyday dynamics that underlie QoL scores and show how specific AT and care arrangements contribute to or detract from lived independence.

2.2 The Social and Emotional Landscape of Dependency: Reframing Autonomy and Care

Mainstream AT discourse often presents independence as a binary state—freedom from reliance on others. Disability studies and feminist ethics critique this framing, advancing relational autonomy, which understands agency as situated within networks of interdependence rather than in isolation [18, 19]. From this perspective, technologies and caregiving relationships do not diminish autonomy; they can constitute its foundation by enabling people to act in line with their values [5, 20, 21]. Our findings support this view, showing that powered wheelchair users experience independence as fluid, continually negotiated, and shaped by daily caregiving dynamics.

This framing also foregrounds the role of emotional labor in care ecologies. HCI scholarship shows caregiving is not merely functional support but an emotionally charged practice involving the regulation of feelings to sustain relationships [9, 47, 58, 70]. Technologies can alleviate routine burdens but may also generate new strains by disrupting rhythms or feeling intrusive [71]. Caregivers often value tools that automate transactional tasks, such as scheduling or retrieving objects, while preserving emotionally meaningful interactions. This resonates with our participants' emphasis on reducing frequent “ask-points,” where dependency feels most acute. Technologies that ignore these affective dynamics risk undermining the very relationships they intend to support.

Attention to dignity, privacy, and stigma is therefore critical. Privacy, understood as control over one's body, space, and information, is central to a dignified care experience [46, 78]. Nevertheless, intimate caregiving tasks often expose vulnerabilities that technology may either mitigate or exacerbate. Similarly, stigma attached to visible assistive devices can produce feelings of conspicuousness, shame, or social anxiety, sometimes leading to device abandonment [30, 34, 54]. Users may face trade-offs between functionality and perceived normalcy. In this light, stigma and dignity are not peripheral “soft” concerns but core determinants of whether technologies are embraced or rejected.

This work, thus, situates independence not as separation from others but as the capacity to navigate dependencies while preserving dignity, privacy, and emotional well-being. Our study extends these insights by showing how powered wheelchair users frame autonomy through both practical and emotional negotiations with caregivers and technologies.

2.3 Navigating the “Advice Maze”: Polycentric Decision-Making in AT Adoption

AT procurement is often depicted as a clinician-led process, where occupational therapists (OT) and physical therapists (PT) assess needs, recommend devices, and navigate insurance systems. In practice, however, decision-making is polycentric and distributed across multiple actors. Clinicians remain important gatekeepers, but their recommendations are not always decisive. Studies show that when clinical advice conflicts with users' lived contexts, decisional conflict emerges, often prompting people to seek additional sources of guidance [13].

Families frequently play a decisive role, filtering technology options through the lenses of daily routines, financial constraints, and caregiving capacity. A family-centered approach is widely recognized as essential, since successful integration depends on how well a device fits within household ecologies [53]. By weighing practical feasibility alongside emotional and relational considerations, families act as pragmatic arbiters of adoption.

Equally important are peer support networks. Both in-person communities and online forums provide experiential knowledge that users often regard as more relevant than formal clinical expertise [31]. Peer-led initiatives, such as Australia's AT Chat, demonstrate how structured sharing of lived experiences can build confidence and improve decision-making [42]. These communities do not simply supplement clinical input; they reflect a deeper epistemological tension between codified professional knowledge and the situated expertise of lived experience. Despite their value, peer perspectives are often undervalued in institutional systems that privilege clinical authority, particularly in funding and reimbursement processes [23].

Grassroots DIY and maker communities represent an emerging source of AT innovation. Platforms like Makers Making Change enable co-creation of customized, affordable devices, shifting problem-solving into the hands of users and community volunteers [44, 49, 51]. These efforts illustrate a decentralized model of innovation where solutions are generated in direct response to lived needs. Beyond producing devices, these communities generate new forms of knowledge that combine technical skill with deep situational awareness.

Thus, these overlapping influences form an “*advice maze*” in which users navigate clinical authority, family pragmatism, peer wisdom, and DIY ingenuity. Rather than signaling dysfunction, this polycentric process highlights the richness of AT expertise and underscores the need for infrastructures, such as our proposed *Assistive-Tech Commons*, that can integrate and elevate these diverse forms of knowledge.

2.4 Interview-based studies with powered-wheelchair users

A focused thread of HCI work uses interviews with powered-wheelchair users (PWUs) to shape both problem framing and method. Early work by Carrington et al. [10] employed formative interviews to reconceptualize the wheelchair as an interactive platform, foregrounding reach, placement, and “chair real estate” as primary design materials rather than treating phones and tablets as external artifacts to be adapted. Building on this, Rodger et al. [63] coupled a go-along method with a participant-operated capture probe to conduct data-led follow-up interviews, elevating participants to co-analysts of in-situ mobility barriers and making visible the otherwise hidden labor and psycho-emotional impacts of access negotiations. More recently, Zhang et al. [83] combined semi-structured interviews with a virtual co-design workshop to articulate internal (user–chair) and external (chair–crowd) interaction loops, reframing shared-control navigation in crowds as a socio-technical challenge that must be socially legible, not merely collision-free. Complementing these, Shahria et al. [65] center on how assistive technology and caregiver support are entangled in everyday life, drawing attention to autonomy, dignity, and structural constraints around care.

These studies demonstrate how interview methods can reconfigure the design space (wheelchair as platform), strengthen methodology through in-situ capture and participant co-analysis, and reorient objectives from raw autonomy to socially legible autonomy. At the same time, they remain largely device-, probe-, or scenario-bounded, with limited instrumentation of day-long trajectories and caregiver ecologies and few pathways for translating qualitative insights into repeatable practice. Our approach extends this lineage by organizing interviews around routine “ask-points” across morning/evening trajectories, explicitly modeling caregiver availability and boundaries, and coupling interviewing with co-design and follow-through procedures that yield actionable, maintainable interventions.

2.5 Toward Sustainable, Justice-Oriented Ecosystems

The limitations of traditional AT models and the recognition of socially embedded use have driven a “participatory turn” in HCI and accessibility research. Participatory Design (PD) reframes disabled people as co-designers and experts in their own lives [3, 56]. Community-Based Participatory Research (CBPR) and Participatory Action Research (PAR) extend this ethos by emphasizing empowerment and structural change, challenging researchers to work with rather than on marginalized communities [15, 16, 26, 43, 74]. While these approaches are often discussed alongside user-centered design (UCD), they differ in important ways from the dominant UCD paradigm in HCI. Classic UCD typically focuses on understanding individual users’ experiences and needs, with designers and researchers retaining primary control over how data are interpreted and translated into design decisions. In contrast, Participatory Design (PD) explicitly treats users and other stakeholders as co-designers and co-creators of knowledge, emphasizing that design outcomes emerge from negotiations among multiple actors

whose viewpoints may overlap, diverge, or even conflict [62]. Similarly, CBPR and PAR foreground shared ownership of the research process, including the co-creation of data, joint analysis, and collective decision-making about actions and interventions. Rather than merely informing a predefined design process, participants help shape research questions, methods, and design directions, shifting power from expert-led to genuinely collaborative, multi-stakeholder practices. Disability justice frameworks go further, calling for intersectional approaches and leadership by those most impacted [17].

These approaches have reshaped the ethos of AT design, but face a persistent limitation: their impact is often confined to time-bound projects. While grassroots maker communities [39, 51] and peer-support networks [28] demonstrate the potential of community-driven innovation, they remain fragmented and disconnected from clinical and funding systems. The gap between participatory ideals and durable infrastructures means that promising solutions frequently fail to persist beyond individual research initiatives [7, 25].

Addressing this challenge requires moving from episodic co-design to sustainable ecosystems. Our proposed Trajectory- and Ask-Point-Centered Co-Design Framework responds by integrating a longitudinal perspective, inspired by patient journey mapping [35, 60]. By focusing on daily trajectories and identifying high-frequency “ask-points,” such as morning routines or door operation, this framework situates design within the lived oscillations of autonomy and dependency. Rather than assessing devices in isolation, it highlights when and where technologies can deliver disproportionate gains in independence and dignity.

Complementing this methodology, the Assistive-Tech Commons is envisioned as a community-governed knowledge infrastructure. Unlike traditional product review platforms, it indexes solutions by ask-point and lived context, integrates DIY modifications, and consolidates funding strategies and real-world evidence. In doing so, it elevates peer and family expertise into dialogue with clinicians and manufacturers. This Commons formalizes fragmented community knowledge into a collective resource, bridging the gap between participatory design principles and long-term sustainability in AT ecosystems.

3 Methods

We conducted a qualitative study with PWUs, combining semi-structured interviews and thematic analysis to gain a comprehensive understanding of the lived experiences surrounding autonomy, caregiving, and technology use.

3.1 Participants

We conducted in-depth qualitative interviews with 55 powered-wheelchair users (PWUs) across ten U.S. states, intentionally selected to capture diverse lived experiences, medical conditions, and caregiving arrangements. Eligibility required current powered-wheelchair use. Participants ranged in age from 18 to 65 years. Conditions included spinal cord injury ($n=7$), quadriplegia ($n=4$), multiple sclerosis ($n=2$), spinal muscular atrophy ($n=2$), and cerebral palsy ($n=2$), among others (Table 1). In addition, the remaining participants reported other mobility-affecting impairments or combinations of conditions, enriching the diversity of our dataset. Thus,

this heterogeneity enabled a nuanced exploration of how different impairments uniquely shape emotional and practical experiences with assistive technologies.

Characteristic	Distribution
Demographics	
Age (years)	18–65
State of residence	FL, GA, IL, MO, ND, NY, OH, PA, TX, WI
Living arrangement	With family (52); Residential facility (3)
Primary condition	SCI (7); Quadriplegia (4); MS (2); SMA (2); CP (2); Other/Not Reported (38)
Care context	Family caregiver (33); Paid caregiver (19); Hybrid (3)
Study logistics	
Interview modality	Remote (44); In-person (11)
Recruitment channel	Healthcare centers; Social media; Abilities Expos (Chicago, Houston)

Table 1: Overview of study participants (n=55), including demographics (voluntary disclosure), living and caregiving arrangements, and study logistics (interview modality and recruitment channel).

Recruitment occurred through multiple channels, including healthcare centers, social media, and Abilities Expos, where powered-wheelchair users often explore new technologies. This approach broadened our reach and allowed us to engage participants with diverse caregiving contexts and levels of technological exposure. Interviews were conducted between July and December 2024, with forty-four held remotely via video conferencing platforms and eleven conducted in person, depending on participants' preferences and accessibility needs.

3.2 Interview Procedure

Interviews followed a semi-structured protocol, consisting primarily of open-ended questions to elicit rich, first-person narratives and to facilitate discussion of sensitive and complex experiences. The interview guide was organized around participants' daily trajectories and high-frequency "ask-points" (e.g., morning routines, toileting, meal preparation, door operation), caregiving arrangements, emotional experiences related to dependency, decision-making around assistive technologies, and systemic barriers and enablers to technology use. Table 2 summarizes sample questions for each topical area, and Appendix A provides the full semi-structured interview guide. To encourage deeper reflection, interviewers employed follow-up probes such as, "Could you tell me more about that experience?" or "What emotions or thoughts arose for you in that moment?" This approach enabled conversational, participant-led dialogue, allowing participants to freely articulate their lived realities, express nuanced emotions, and describe the personal strategies and barriers they encountered in their everyday lives.

During the conversation, we intentionally did not ask participants to disclose demographic details such as age, gender, years of wheelchair use, or specific diagnoses. These characteristics were only noted when participants chose to self-disclose, to minimize

potential discomfort or feelings of medical scrutiny. Interviews lasted on average 45 minutes (range: 30–90 minutes), were audio-recorded with participant consent, and transcribed verbatim for analysis. Data collection proceeded until thematic saturation was reached, with no new major themes emerging after approximately 50 interviews. An additional five interviews were conducted to confirm saturation and capture variation across different lived contexts.

3.3 Data Analysis

We employed thematic analysis following Braun and Clarke's rigorous qualitative approach [6]. This was an inductive analytic process, allowing patterns and themes to emerge directly from participants' narratives rather than being shaped by pre-existing theoretical constructs. We adopted a transparent, hand-coded workflow using shared spreadsheets and a coding workbook to facilitate collaborative iteration. Transcripts underwent iterative review to identify preliminary patterns and develop an initial list of themes. As analysis progressed, we iteratively revised the codebook: when segments did not fit existing codes or revealed a distinct pattern, we added/updated codes and revisited earlier transcripts to maintain consistency until themes stabilized. Two researchers then independently performed initial inductive coding, generating codes directly grounded in participants' own words and experiences and recording emerging reflections in memo fields. Since the interviews followed a semi-structured format and focused on specific topics, the coding process primarily organized responses around individual or groups of related topics. Inter-rater agreement for this phase was substantial ($\kappa = 0.78$); discrepancies were discussed collaboratively, leading to refinement of code definitions and the consolidation of overlapping codes. We then clustered related codes into higher-level categories that became the Findings themes (Section 4), with subthemes reflecting recurring code clusters; Table 3 provides traceability from themes/subcodes to exemplar excerpts.

The research team included members experienced in disability studies and assistive technology design, as well as those newer to the domain. This diversity facilitated ongoing dialogue and rigorous reflection on potential biases throughout data analysis, strengthening the trustworthiness and interpretive rigor of our findings. Periodic peer-debriefing sessions further ensured interpretive rigor, transparency, and methodological robustness, aligning our findings with participants' lived realities and voices.

3.4 Ethical Considerations and Positionality

This research was approved by our Institutional Review Board. All participants provided informed consent, with procedures presented in accessible formats. We emphasized that participation was voluntary and that individuals could decline to answer any question or stop the interview at any point without consequence. Because conversations often touched on sensitive issues such as privacy, dependence, or caregiving, interviewers were attentive to signs of discomfort and offered breaks or redirection as needed. To ensure equitable access, participants could choose between remote or in-person interviews, and accommodations such as flexible scheduling and support for alternative communication methods were made available. All interviews were audio-recorded only with

Topical area	Example question	Related RQ(s)
Daily routines and ask-points	Can you walk me through a typical morning in your day, from when you wake up until you feel ready to start your day? As you describe it, could you point out any moments when you usually need to ask someone for help?	RQ, RQ1, RQ2
Caregiving arrangements and relationships	Who helps you with your day-to-day activities, for example, family members, friends, or paid caregivers? How does that support usually work across a typical day?	RQ, RQ1
Emotional experiences of dependency	Can you share an example of a time when relying on others for help felt emotionally difficult or uncomfortable?	RQ, RQ1
Technology use in daily life	What kinds of assistive technologies do you currently use in your daily life? What other technologies, like smartphones or low-tech tools, do you use that help you manage your day?	RQ, RQ1, RQ2
Decision-making and advice networks	Thinking about a specific device you use a lot, can you tell me the story of how you started using it? Who first suggested it to you? How do you usually gather information?	RQ, RQ2, RQ3
Systemic barriers and alternatives	Have there been times when insurance, funding, or other policies made it hard to get, keep, or repair a device you wanted? How did you handle that situation?	RQ, RQ2, RQ3

Table 2: Sample semi-structured interview questions by topical area.

consent, transcribed verbatim, de-identified, and stored securely. Participants received a \$30 gift card as compensation for their time.

Our positionality as a research team informed both the design and interpretation of this study. Two members have close family connections to disability, including long-term caregiving for relatives. These personal experiences provided insight into the complexities of dependency, dignity, and day-to-day negotiations of care. Other members bring professional expertise in assistive and rehabilitation technologies, human-computer interaction, and accessibility research. These perspectives enriched our sensitivity when engaging with participants and shaped our ability to recognize the layered emotional and practical challenges raised in interviews. At the same time, we remained mindful that our backgrounds could influence interpretation. To address this, we employed collaborative coding, conducted frequent peer debriefings, and committed to centering participants' narratives as checks against our assumptions.

4 Findings

The goal of our findings is to illuminate how powered-wheelchair users (PWUs) experience and negotiate independence in their daily lives, and how caregiving arrangements and technologies intersect with these negotiations. Our analysis is based on 55 in-depth interviews with PWUs, whose situations span a broad assistance spectrum, from 1 to 24 hours of daily help (with an average of 8.7 hours), with caregiving arrangements predominantly being family-based ($n=33$), paid-caregiver dependent ($n=19$), or hybrid ($n=3$). Tasks most commonly triggering external assistance centered around (1) fine-motor reach tasks (picking up objects, opening containers), (2) personal care (toileting, showering, dressing), (3) meal preparation, and (4) environmental access (doors, transfers). Across these situations, participants frequently described assistive technologies

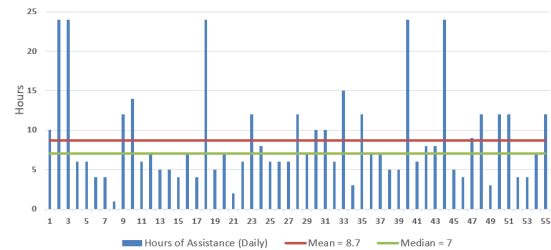


Figure 1: Daily hours of hands-on caregiver assistance reported by participants ($n=55$). The red line indicates the mean (8.7 hours/day), and the green line indicates the median (7 hours/day). The distribution is right-skewed, with several high-dependency outliers requiring near round-the-clock support.

as deeply empowering yet ultimately inadequate to remove many high-frequency “ask-points” entirely:

“If I have to ask somebody to help me with the device more than I can use it independently, then it’s not that helpful.” (P2)

“I have my ceiling lift for transfer. But I have a caregiver who comes to my home and starts my morning routine, which includes getting cleaned up, dressed, and then they use the ceiling lift to put me into my chair.” (P15)

Building from these, the findings unfold across four interconnected themes: *lived realities and independence, the social and emotional impact of caregiver dependency, decision-making in technology and care, and the barriers and enablers of technology adoption*. Each

Theme	Subcode	Exemplar quote
Lived Realities and Independence	Tech amplifiers of agency: Robotic arm enabling solo time	<i>"Before the JACO arm, it was very hard for me to be alone at any time. Now I can be left alone for several hours." (P26)</i>
	Subjective independence: Independent in tasks, not chores	<i>"I'm pretty independent, more than probably most people, as far as accomplishing daily tasks." (P8)</i>
Social and Emotional Impact of Caregiver Dependency	Privacy with paid care: Boundary-setting takes time	<i>"At the beginning, I felt my caregiver was invading my privacy, but with time I got to understand and she keeps boundaries now." (P10)</i>
	Care delays & health: Delays can trigger health issues	<i>"So I had to wait till she got back, and I was so hungry it triggered my process. That gave me a health issue for about a week or two." (P47)</i>
	Emotional labour and loneliness: delayed / missing care	<i>"I need assistance. Most of the time... I find it difficult to move around and do some stuff. And also, I feel lonely most of the times." (P34)</i>
Decision-Making in Technology and Care	Clinical expertise: Therapists consulted first	<i>"Most of the time I talk to therapists because they're more knowledgeable." (P24)</i>
Barriers and Enablers of Technology Adoption	Cost & coverage: Out-of-pocket burden	<i>"The amount of benefits you get from insurance is very limited. You have to pay out of pocket for most things." (P17)</i>
	Caregiver shortages: Availability & trust gaps	<i>"Finding good caregivers, especially in California, is very, very hard." (P45)</i>
	Policy and accessibility friction: misfit between technology and environment	<i>"My workplace is also not accessible in areas. I need help a lot when I have to go to work. The office is upstairs and I need someone to help with that." (P51)</i>

Table 3: Traceability between major themes, subcodes, and exemplar participant quotes from interviews. Themes map onto the overarching RQ and sub-questions (RQ1–RQ3), spanning lived experiences of independence, decision-making around technology and care, and structural barriers and enablers.

theme highlights different dimensions of how PWUs negotiate autonomy, dependency, and the role of assistive technologies within everyday contexts. Table 3 shows the recurring themes, corresponding sub-themes and example quotes.

4.1 Lived Realities and Independence

Participants' daily caregiving needs varied substantially. The distribution was right-skewed (Fig. 1) with five participants reporting receiving near round-the-clock support, creating visible outliers, while the majority clustered well below ten hours. The mean was 8.7 hours per day, slightly above the median of seven, reflecting the influence of these high-dependency cases. Roughly one-third of participants managed with fewer than six hours, suggesting a segment with intermittent but not continuous reliance. Others showed mid-range needs between eight and twelve hours, often associated with multiple assistance tasks spread across the day.

Table 4 reveals that primary dependency was concentrated around a limited set of recurring activities. Picking up or reaching objects dominated, with 87% of participants requiring help, underscoring how seemingly minor reach tasks compound into broader dependence. Food preparation (67%) and personal hygiene (62%) emerged as the following most challenging domains, both central to daily routines and emotional well-being. More than half also identified opening doors as a barrier to independence (53%), illustrating how environmental design can amplify dependency. Other tasks, such as dressing (40%), transfers (33%), and household

chores (25%), were less frequent but represented consistent points of vulnerability. As one participant reflected, *"I can't shower or cook for myself, and picking things up from the floor or opening doors is very hard for me."* (P20). This illustrates how dependencies often spanned across multiple domains, compounding routine challenges. The overlap across these domains suggests that a relatively narrow set of tasks drives much of the caregiving burden.

Despite this quantitative variability, participants often assessed technology and routines based on their ability to meaningfully reduce moments requiring external help (referred to as "ask-points"). Failures in these tasks could cascade into broader disruptions, intensifying the sense of dependency. Technologies perceived as effective were those specifically reducing these frequent "ask-points," transforming minor task autonomy into substantial gains in overall independence. Within this theme, two sub-themes emerged: how specific technologies amplified participants' agency, and how independence itself was redefined through everyday negotiations of privacy, safety, and relational comfort.

4.1.1 Technological Amplifiers of Agency. Participants' accounts revealed that certain technologies acted as powerful amplifiers of agency, reshaping their capacity to navigate daily life with greater independence. They described certain technologies as transformative, enabling previously unattainable levels of independence and autonomy. The Kinova Jaco robotic arm [38] emerged as particularly empowering, as illustrated by participants who reported

Activity	# of Participants
Picking up / reaching objects	48 (87%)
Food preparation or eating	37 (67%)
Personal hygiene (shower/toilet)	34 (62%)
Opening or operating doors	29 (53%)
Dressing and grooming	22 (40%)
Transfers in / out of bed	18 (33%)
Household cleaning / laundry	14 (25%)

Table 4: Most frequently cited daily activities that required external assistance among participants (n=55). Percentages reflect the proportion of participants who reported needing help with each task; responses are not mutually exclusive.

significant gains in autonomy. For example, P26 reported *being able to spend several hours alone after adopting a JACO robotic arm*.

Similarly, another participant (P28) reported tolerating five caregiver-free nights weekly because the robotic arm enabled tasks such as lifting glasses, picking up dropped items, and interacting with personal stuff independently. Such advanced technology improved their quality of life, but very few participants explored these options. Only 3 of 55 participants reported using a robotic arm (5%), while seven others were merely aware of them, despite frequent reports of reach-related dependency. Cost and limited insurance coverage drove this gap. Among the three JACO users, only one arm was insurance-funded, with numerous hurdles; the other two were acquired through self-payment or crowdfunding. A high up-front price, limited trialability, and uncertainty about maintenance and repair can deter adoption, even when interest is substantial. Weight, mounting constraints, and battery draw introduce practical trade-offs; users weigh these against the reliability of human help and low-tech workarounds.

In addition to advanced robotics, participants highlighted that everyday and often overlooked technologies also amplified their agency. Adoption clustered around tools that are inexpensive, easy to obtain without prescriptions, and simple to replace when damaged. Voice assistance on smartphones topped the list (24%), followed by smart-home assistants (18%) and reacher/grabber devices (18%). These options operate within ecosystems that participants already own or can purchase off the shelf without any insurance approvals, evaluations, or delivery delays.

“When choosing a new device, my first concern is usually price. Secondly, how easy it is to get and use, like user-friendly. (P47)”

Therefore, breakage is a nuisance rather than a crisis; items can be swapped quickly, often for <\$50 in the case of reachers. Moreover, mainstream voice tools also help avoid medical stigma and require minimal training, which reduces the risk of abandonment. However, participants also highlighted the limits of such low-cost tools. For instance, one explained,

“I usually carry a grabber around with me... the grabber isn’t perfect if an object is too heavy or oddly shaped. If something falls and I can’t pick it up, either it lives on the floor until somebody comes over or I have to get down and get it.” (P39)

Technology	# of Participants
Voice assistance (smartphone)	14 (24%)
Voice assistance (smart-home; Google/Alexa)	10 (18%)
Reacher / Grabber	10 (18%)
Automated / electric door opener	7 (13%)
Custom object / 3D-printed object	3 (5%)
Robotic arm (JACO)	3 (5%)
Others (Health monitoring device, feeding device)	3 (5%)

Table 5: Assistive and mainstream technologies used by participants (n=55). Adoption concentrated on low-cost, readily available tools, whereas advanced technologies appeared infrequently, primarily constrained by cost and limited access. Responses are not mutually exclusive.

This underscores how even widely adopted solutions often provide only partial independence. Automated door openers appeared less frequently (13%). The benefit is high, as door operation is a persistent ask-point; however, installation costs, landlord permissions, and home wiring complicate access. Custom or 3D-printed objects (5%) signal maker activity and peer knowledge, but require design time, printer access, or a helpful community member, which constrains scale.

As summarized in Table 5, participants gravitated to technologies that target high-frequency ask-points while sidestepping insurance gatekeeping, complex onboarding, and downtime risk. This theme, therefore, combines a broadly shared pattern (mainstream/low-cost tools used by many participants) with a high-impact but less common experience (advanced robotics, such as JACO, reported by a small minority, n=3). High-impact technologies remained rare not because the need is low, but because of acquisition, serviceability, awareness, and fit within home and funding ecologies create a steep adoption threshold. Overall, these patterns underscore how both advanced robotics and everyday mainstream tools can significantly amplify agency, though systemic and practical barriers continue to limit widespread uptake.

4.1.2 Redefining “Independence”. Beyond devices themselves, participants reflected deeply on what “independence” meant in practice, revealing that it was often defined in more complex and personal terms. For many participants, the notion of independence extended beyond the mere reduction of caregiving hours to encompass emotional and psychological well-being. Independence was often understood as subjective, negotiated within the context of daily interactions and support systems. Some participants self-described as independent for most daily tasks yet still relied on others for heavier chores (see Table 3), thereby highlighting the complex negotiation inherent in defining independence.

“I’ve had to schedule my life around when the caregiver could come; sometimes that meant delaying plans or taking a break from what I was doing.” (P5)

Within this subjective framing of autonomy, personal privacy emerged as a critical dimension, particularly when daily routines required intimate care from family or paid caregivers. Paid caregivers often presented alternative trade-offs: greater professional boundaries and predictability but less relational intimacy, highlighting a persistent tension between independence, emotional comfort, and privacy. Nonetheless, some participants, such as P16, reported

discontinuing paid caregiving altogether because it felt more intrusive than supportive, ultimately shifting reliance back to family members. Others acknowledged privacy as a concern but described it as a secondary priority compared to safety and independence, suggesting that while privacy remains valued, it does not always take precedence over practical caregiving needs.

These lived realities underscore that independence is not reducible to fewer caregiving hours or advanced devices alone. Instead, it emerges from a layered negotiation across assistive technologies (from reachers to robotic arms), caregiving arrangements (family versus paid), and personal priorities (privacy, safety, or emotional comfort). Independence, in this sense, is best understood as a fluid, relational, and technologically mediated experience rather than a fixed outcome.

4.2 Social and Emotional Impact of Caregiver Dependency

Caregiver dependency shaped participants' lives through practical assistance and profound emotional and relational consequences. In the findings that follow, we highlight two sub-themes: the impact of caregiver dependency on privacy, dignity, and relational dynamics, and the emotional labor and loneliness that often accompanied inconsistent or unavailable care.

4.2.1 Impact of Caregiver Dependency. Building on the privacy concerns identified above, caregiver dependency also carried profound social and emotional ramifications. While participants often expressed gratitude for receiving care, these arrangements simultaneously introduced layers of vulnerability and emotional strain. Dependency was rarely a neutral condition. It was experienced through a constant negotiation between comfort, frustration, and the preservation of dignity. Privacy emerged as a prominent concern among participants, with almost half ($n=26$) explicitly describing anxieties around constant surveillance, emotional exposure, and loss of autonomy, particularly during intimate caregiving tasks.

"The moment I'm awake, everyone's around me 24/7, and for someone who's an introvert. That's like abuse of privacy." (P32)

Family caregiving relationships, though frequently rooted in trust and emotional closeness, were not immune to conflict. Instead, participants noted that intimacy often intensified role ambiguity, producing feelings of infantilization and frustration. For example, participants mentioned,

"It's like being an adult and still being babysat." (P32)
"I struggle with the bathroom most especially, it's really challenging. It makes me feel very vulnerable, and that's not good. Especially when my husband is not around." (P51)

In contrast, paid caregivers typically brought clearer professional boundaries and more predictable routines. However, their involvement often generated an initial sense of discomfort because of the unfamiliarity and emotional detachment of having strangers in intimate personal spaces. Others initially experienced paid care as privacy-invasive, but the boundaries eventually settled as one participant explained,

Affective theme	# of participants
Privacy invasion	26 (47%)
Feeling like a burden	20 (36%)
Loneliness / isolation	15 (27%)
Emotional safety	7 (13%)
Infantilization	4 (7%)

Table 6: Dominant affective themes surrounding caregiver dependency, coded from participant narratives ($n=55$). Percentages represent the proportion of participants who explicitly referenced each theme.

"At the beginning, I felt my caregiver was invading my privacy, but with time I got to understand and she keeps boundaries now." (P10)

These reflections highlight how both family and paid arrangements offered distinct advantages and challenges. In either case, dependency did not simply involve the practical completion of tasks. It reshaped emotional landscapes, exposing tensions between comfort, autonomy, and relational balance. Beyond these relational dynamics, participants emphasized that dependency itself created an ongoing form of emotional labor, especially when caregiving was inconsistent or unavailable.

4.2.2 Emotional Labour and Loneliness. Caregiver shortages intensified these emotional costs, leaving participants vulnerable to gaps in essential routines. Delays in care were not merely inconvenient but carried direct health and psychological consequences. Participants also described moments of acute vulnerability when care was unavailable, which highlights how inconsistent caregiving arrangements disrupted routines and created feelings of helplessness and social dependence. As one explained,

"Sometimes, when my husband isn't around, I feel almost useless for certain things. I have to call neighbors if it's urgent." (P51)

Delays sometimes had direct health repercussions, as waiting for a long time can trigger complications (see Table 3). Experiences like these illustrate how dependency extended beyond functional support to encompass profound emotional burdens.

"I needed to refill my bottle, so I had to wait until my mom or anyone else came by and gave me water. I felt so helpless." (P41)

Feelings of helplessness, loneliness, and frustration often surfaced when care was delayed or absent, highlighting the fragility of reliance on external support. Within this context, independence was defined as the ability to perform physical tasks and as a source of emotional relief. Even the automation of a single routine, such as operating a door or retrieving an object, was described as significantly reducing the psychological weight of constant dependency. In this sense, technologies that mitigated "ask-points" acted as emotional stabilizers, offering participants a respite from the strain of continual reliance.

These accounts reveal that caregiver dependency was never experienced in isolation from its emotional consequences. Family and paid caregivers both shaped participants' sense of dignity and

autonomy in distinct but complex ways, while shortages or delays in care heightened vulnerability and distress. Thus, emotional and social dimensions profoundly shaped participants' caregiving experiences and attitudes towards technology. These emotional realities directly influenced how PWUs navigated decisions around technology and caregiving arrangements, a point we examine next.

4.3 Decision-Making in Technology and Care

Emotionally laden experiences profoundly impacted how PWUs navigated the complex “advice maze” (P8) surrounding technology choices, as they balanced formal expertise, informal support, and personal intuition. Participants described their technology choices as navigating an often-confusing landscape characterized by multiple competing voices and influences. While clinicians were central in initial technology procurement conversations for roughly a third of participants (n=18), though 40 engaged with OTs/PTs at some stage, the overall decision-making process often integrated informal networks, such as family members, self-driven research, and peer communities, which collectively held greater sway than clinical expertise alone. We identified three recurring sub-themes in participants' accounts of decision-making: the tensions between clinical authority and lived expertise, the pragmatic role of families, and the creative influence of peer and DIY networks.

4.3.1 Clinician Gatekeeping versus Lived Expertise. Participants frequently described clinicians as both indispensable and limited. On one hand, therapists were valued for their ability to navigate bureaucratic barriers such as insurance documentation, equipment evaluations, and clinical justifications. This positioned them as the formal “gatekeepers” of technology adoption, often serving as the first point of contact in securing devices. As one participant noted, “Most of the time I talk to therapists because they're more knowledgeable.” (P24) Several participants acknowledged therapists' specialized knowledge and appreciated their advocacy within complex healthcare systems. However, these same gatekeeping functions also highlighted critical shortcomings. Recommendations were frequently perceived as overly generic, failing to capture the lived nuances of everyday life with a wheelchair. Participants noted that therapists often prescribed only the most essential or traditional devices, such as the wheelchair itself, while overlooking the range of assistive technologies that might reduce daily ask-points:

“My therapist never recommended any assistive technology besides the wheelchair. The therapist should be the one to recommend technology.” (P38)

“Most of the house is not built for a wheelchair-abled person... So, when I'm in my chair, I have to be innovative myself.” (P46)

This gap created frustration, as participants felt their expertise as daily users was undervalued. Some described moments where their intimate knowledge of routines, privacy needs, or home environments directly conflicted with clinical advice. The consequence was a persistent tension between professional authority and lived expertise. As one participant explained, “Therapists know the system and they have more experience than others, but their suggestions are limited most of the time” (P10). This sentiment captured a recurring

tension: clinical expertise could unlock access, but it often failed to align with the texture of everyday life fully.

Importantly, these accounts underscore a broader pattern in which professional expertise provides access but not necessarily alignment. Clinicians were seen as effective at initiating the process of procurement, but ineffective at ensuring long-term integration. In practice, this often shifted decision-making authority away from the clinic and toward families or peer networks, where contextual knowledge could be more fully recognized and acted upon.

4.3.2 Family Pragmatism. While clinicians often initiated formal adoption processes, families played a decisive role in shaping whether technologies were pursued, integrated, or abandoned. Participants emphasized that families approached decisions pragmatically, weighing technological possibilities against household routines, caregiving capacity, financial constraints, and emotional well-being. Unlike clinicians, whose judgments were grounded in professional standards of “medical necessity,” family members evaluated technologies through the lens of sustainability and everyday fit:

“If there's a new technology that would help, especially to reduce the burden on my wife, I'd love to go for it.” (P16)

This framing positions assistive technologies not as individual tools but as shared household investments. Participants described how family members often prioritized solutions that reduced caregiver fatigue, enabled smoother household rhythms, or minimized long-term costs. Participants frequently framed technology decisions in terms of easing the demands of family caregiving. One noted,

“I think if I can actually get a device that will be able to do some tasks for me... it would be very, very nice because there are some tasks that I would actually love to be done without any stress for my wife.” (P23)

Such reflections illustrate how family pragmatism often translates into a desire for technologies that alleviate relational strain as much as they support individual independence. Decisions were thus rarely about independence in isolation; they were about balancing independence with relational harmony and collective well-being. For some, this meant adopting mainstream, inexpensive devices that could be easily replaced; for others, it meant forgoing advanced equipment due to high maintenance demands or the risk of disruption when it failed.

However, this pragmatic stance was double-edged. It ensured that technologies aligned with the lived realities of caregiving ecologies. On the other hand, it sometimes curtailed experimentation, as families tended to minimize risk and avoid devices perceived as financially or emotionally burdensome. Family pragmatism, therefore, functioned as both a safeguard and a gatekeeper, enabling feasible adoption but constraining innovation. This pragmatic logic contrasted sharply with peer networks, which participants described as spaces more open to creativity, customization, and risk-taking. Hence, these dynamics illustrate how decision-making authority was negotiated polycentrically, with family pragmatism often serving as the decisive filter through which clinical advice and peer experimentation were ultimately evaluated.

Channel	# of Participants
Occupational / Physical therapist	40 (73%)
Family or spouse	32 (58%)
Peer PWU networks (online / expos)	18 (33%)
Manufacturer reps / distributors	12 (22%)
Social media reviews	11 (20%)
Crowdfunding campaigns	5 (9%)

Table 7: Information channels participants engaged with during assistive technology adoption decisions. Values indicate the number (and proportion) of participants who reported using each channel; participants often engaged multiple sources.

4.3.3 Peer Wisdom and DIY Innovation. Peer wheelchair-user networks, online forums, and the maker community significantly shaped technology adoption by filling gaps left by clinical and family-centered decision-making. These channels provided first-hand experiences and enabled participants to pursue unconventional or customized solutions:

“With a 3D printer of my own, I’m trying to find things I can build on my own to help myself out.” (P21)

“A lot of things, like my standing frame, I found out about through friends or the community sharing their experiences.” (P8)

Rather than passively receiving expert advice, participants actively sought experiential knowledge from fellow users, leveraging grassroots ingenuity as a complement, and at times an alternative, to formal pathways. Such peer-driven experimentation reinforced the importance of community expertise in validating technologies and adapting them to real-world use (Table 7).

“I am currently not connected with others other than some of my friends... I wish to connect with such networks and learn about their experiences.” (P51)

Assistive technology decision-making emerged as a layered negotiation involving clinical authority, familial pragmatism, and peer-driven innovation. Each perspective provided unique benefits but also distinct limitations, requiring participants to reconcile competing priorities of feasibility, cost, emotional comfort, and creativity. Even when effective decisions were reached, broader structural, economic, and social barriers often constrained actual adoption and sustained use. These dynamics highlight that individual decision-making, while critical, cannot guarantee technological success on its own, a complexity explored further in the following section.

4.4 Barriers and Enablers of Technology Adoption

Technology adoption was rarely a straightforward outcome of individual choice. Participants emphasized that successful integration depended as much on surrounding infrastructures, insurance, caregiver support, training, and environmental access as on the device itself. At the same time, adoption was enabled when social networks, responsive clinicians, or community ingenuity filled systemic gaps.

We next explore these dynamics by outlining the barriers that most often constrained adoption, followed by the enablers that made technology use more sustainable.

4.4.1 Barriers of Technology Adoption. Even when effective technology decisions were made, structural, economic, and social barriers frequently impeded adoption and sustained use, showing that individual choices alone cannot ensure success. Participants emphasized that adoption depended less on device capabilities and more on the readiness and support of the broader caregiving ecosystem. Our analysis identified five prominent barrier clusters:

- (1) **Cost and Insurance Denial** (n=42): Financial barriers emerged as the most significant obstacle, particularly for advanced robotic and environmental control technologies. Insurance coverage criteria prioritized “medical necessity” over lived functionality, making it difficult to obtain approval for devices supporting dignity, privacy, or convenience. As a result, participants turned to crowdfunding, informal lending, or personal savings—strategies that unevenly distributed access depending on socioeconomic status:

“The Kinova Jaco Arm is an exceptional wheelchair attachment, but it’s extremely expensive—about \$60,000. No insurance or foundations help cover it, so you have to fundraise or pay out of pocket.” (P52)

These narratives highlight how financial systems act not as neutral backdrops but as gatekeepers mediating whether technologies can have a meaningful impact.

- (2) **Caregiver Shortages** (n=20): Even the most capable devices could not be fully integrated without consistent caregiver support. The pandemic and ongoing workforce shortages exacerbated these gaps, leaving participants vulnerable to lapses in essential routines and increasing emotional strain:

“One of the hardest things for me is to find people I trust to provide the services that I need.” (P36)

Without reliable support, even well-designed devices risked being underutilized or abandoned.

- (3) **Training and Knowledge Gaps** (n=15): Participants pointed to inadequate onboarding and inconsistent clinical expertise as barriers to effective use. Devices were often introduced without sufficient training or follow-up, leaving individuals to troubleshoot independently or abandon tools that did not fit their routines. In some cases, clinicians themselves lacked up-to-date knowledge:

“My therapist never recommended any assistive technology besides the wheelchair.” (P38)

This highlights the need for stronger ecosystems of technical literacy, peer-to-peer tutorials, and iterative onboarding rather than one-time prescriptions.

- (4) **Design and Reliability Issues** (n=12): Design flaws and poor durability undermined trust and usability. Fragile components and incorrect installations often pushed participants back to human caregivers, negating independence gains:

“Poor design and incorrect installation reduce effectiveness.” (P36)

“That day, before he left the house, he put water, like the bottle of water on the top shelves... I tried as possible as

I held on to one of the counters... it flipped. I fell off my wheelchair.” (P35)

Failures in critical moments, such as transfers, toileting, or food preparation, carried disproportionate emotional and practical costs.

- (5) **Policy and Accessibility Friction** (n=10): Participants highlighted the misalignment between personal technologies and inaccessible infrastructures. Devices could be rendered ineffective by barriers in transportation, housing, or funding systems:

“A lot of times, my frustrations come from general environmental accessibility. Something that’s a big concern for me is transportation.” (P42)

Here, independence was not about devices alone but whether the surrounding environment accommodated them, underscoring the need for systemic, policy-driven interventions alongside device design.

These barriers show that technology adoption is inherently systemic; without comprehensive ecosystem support, even the most promising interventions risk failure.

4.4.2 Enablers of Successful Adoption. Despite these systemic hurdles, participants identified several counterbalancing enablers that facilitated adoption and integration. Peer support networks, through online forums, expos, and informal communities, have emerged as compelling spaces for sharing experiential knowledge, custom, and DIY modifications. Responsive clinicians who collaborated with users, rather than prescribing from a distance, were also cited as enablers. Crowdfunding and community-driven fabrication filled financial and technical gaps, demonstrating how grassroots ingenuity compensates for institutional shortcomings. One participant summarized this unmet need for centralization:

“There’s a gap between manufacturers and users. Everything is so spread out. If there were a single online catalog or database showing all assistive tech and what it can do, that would help people like me find solutions.” (P52)

These enablers highlight that successful adoption arises not from isolated devices but from collective infrastructures of knowledge, support, and trust. They underscore the need for holistic, multi-faceted interventions, such as financial, infrastructural, and social, rather than device enhancements alone. We further elaborate on practical strategies for addressing these systemic issues in the Discussion (Section 5).

5 Discussion

Our analysis suggests that powered-wheelchair use is best understood not simply as a functional task supported by discrete technologies, but as an ongoing negotiation shaped by technological affordances, caregiving relationships, personal agency, and systemic barriers. Drawing on participant narratives (Section 4), we revisit our research questions about (i) how powered-wheelchair users experience independence across their daily trajectories (RQ1), (ii) how caregiving ecologies and technologies jointly configure that independence (RQ2), and (iii) how assistive technologies are selected, adopted, and sustained within this broader system (RQ3). We

articulate implications for design, methods, and policy, including an empirically grounded conceptualization of fluid independence, a systems model of assistive-technology use, and a user-led design framework and Assistive-Tech Commons. While prior disability studies and HCI work have critiqued binary notions of independence, documented emotional dimensions of care, and described maker cultures in disability communities, our contribution is to operationalize these insights as measurable trajectories, design levers, and a concrete, user-led infrastructure. In doing so, we move from conceptual critique to actionable methods and systems that clinicians, policymakers, and communities can adopt.

5.1 Interpretation of the Findings

The findings highlight that powered-wheelchair use cannot be reduced to questions of device functionality alone, but must be interpreted through the interplay of autonomy, caregiving dynamics, emotional well-being, and systemic barriers. In what follows, we unpack these dimensions to clarify how our results extend existing frameworks and surface new opportunities for HCI research and design. Specifically, we surface three interlocking constructs, fluid independence, emotional externalities in caregiving ecologies, and polycentric, participatory decision-making, that operationalize long-standing debates in disability studies and assistive-technology research and directly answer RQ1–RQ3.

5.1.1 Independence as Fluid Negotiation. Contrary to prevailing frameworks that conceptualize independence as a binary or measurable by hours of unsupervised time, our findings underscore independence as a fluid, contextually sensitive negotiation. Even participants who required near-constant assistance described powerful moments of autonomy enabled by specific technological interactions, such as independently pouring a drink with a robotic arm (P26). Conversely, participants with lower overall assistance requirements still articulated acute dependency during intimate or time-sensitive tasks, as illustrated vividly by P47’s account of delayed eating triggering severe health repercussions.

This fluid and nuanced conceptualization suggests that HCI researchers and policymakers should shift away from rigid independence metrics toward dynamic, *trajectory-based measures* that reflect users’ daily oscillations between autonomy and dependence. Echoing disability studies’ advocacy for “temporal analytics of care,” technologies should be evaluated explicitly under conditions of *peak dependence* rather than only routine scenarios, thereby better aligning device design with lived experience. Practically, this highlights the importance of designers explicitly considering and evaluating devices under conditions of maximum dependence, such as morning or evening routines, when assistive technologies can have the most significant impact on user experiences. This redefinition of independence as fluid and contextually shifting extends critiques of static models, such as HAAT and PEO, while empirically grounding the relational autonomy perspective advanced in disability studies, which frames autonomy as continually negotiated through interdependence rather than as a binary, detached state. Where scholars have developed these arguments conceptually [4, 66], our study contributes a measurement and design toolkit, the ask-point index and privacy-gradient mapping (Section 5.2), which provide concrete

ways to quantify oscillations in autonomy and to target design interventions at moments of peak dependence. We do so by (i) tracing how independence oscillates across high-frequency “ask-points” in daily life and (ii) linking those oscillations to concrete design levers, such as door openers, reachers, or robotic arms, that measurably compress waiting, negotiation, and emotional burden.

5.1.2 Emotional Externalities in Caregiving Ecologies. Our data extend prior research on emotional labor in caregiving [24] by highlighting specific emotional externalities, particularly privacy invasion, infantilization, and vulnerability, as critical but often hidden costs of caregiving. Participants clearly articulated these affective dimensions as barriers to technology adoption, particularly in contexts where assistive technologies exacerbate these emotional stressors (e.g., voice assistants in intimate caregiving moments). Nearly half of our participants explicitly described privacy invasions, over a third reported feeling like a burden, and more than a quarter discussed loneliness (Table 6), underscoring that these emotional costs are not peripheral but central to how assistive technologies are evaluated in practice.

“No, I don’t use Siri. I don’t like her listening to me all the time. Also, she doesn’t understand my English so well. So, it’s kind of frustrating.” (P55)

Family caregiving, while rooted in relational trust, often blurred boundaries, leading participants to describe their experiences as “being an adult and still being babysat” (P32). Conversely, paid caregivers introduced professional boundaries but triggered privacy invasions (“invading my privacy”, P10) and emotional discomfort:

“There’s just something about emotions that paid caregivers miss.” (P17)

Recognizing emotional externalities underscores the necessity of designing technologies and caregiving arrangements that explicitly protect emotional safety, privacy, and dignity alongside functional support. Designers must explicitly incorporate privacy-sensitive features, such as adjustable user-controlled privacy modes or context-aware interfaces, to safeguard emotional dignity during intimate caregiving moments. These findings resonate with HCI scholarship on emotional labor in care, which emphasizes that technologies can either alleviate or exacerbate emotional strain, and with disability studies that identify privacy and stigma as primary determinants of device abandonment. Our participants’ accounts extend this literature by demonstrating how even small ask-points carry disproportionate emotional weight. This moves beyond prior accounts of stigma and embarrassment around conspicuous devices by showing how micro-interactions, such as a single request for help in the bathroom or kitchen, can dominate people’s emotional ‘budget’ for technology use and thereby determine long-term adoption. In contrast to work that treats these emotions as downstream consequences, our findings show that emotional safety, privacy, and dignity function as first-order design constraints that determine whether technologies are adopted or retained at all.

5.1.3 Polycentric, Participatory Decision-Making. Our findings challenge traditional assistive technology procurement pipelines, which typically center on clinical authority. Instead, participants navigated a *polycentric advice maze*, highlighting the importance of peer wheelchair-user networks, DIY maker communities, and

family-driven pragmatism. Participants’ use of 3D-printed parts, low-cost hacks, and crowdfunding campaigns exemplifies frugal, situated innovation that resonates with work on jugaad-inspired approaches [59, 75], socio-cultural design of assistive technologies, but here these practices emerge as mainstream, not marginal, routes for solving everyday problems. Decisions were often finalized outside professional channels, especially when initial clinical recommendations failed to align with lived experience or when insurance denials necessitated alternative pathways (e.g., crowdfunding, P26).

“I got only my powered wheelchair. But as you know, it does not give me much flexibility to make my life easier. So, after getting some ideas from my friends and some online searches, I took a proactive approach to customizing the wheelchair. I used a 3D printer to create various add-ons that enhance its functionality and better meet my needs.” (P12)

This aligns strongly with contemporary design-justice frameworks [17] advocating community expertise over paternalistic clinical authority. Operationalizing these insights implies deliberately integrating community-driven knowledge repositories, accessible modding toolkits, and formalizing crowdfunding as legitimate financing mechanisms within mainstream assistive technology ecosystems. Implementing this approach requires the systematic inclusion of community-curated reviews, DIY modification databases, and integrated crowdfunding tools directly within technology procurement platforms. In line with prior work on the ‘advice maze’ of assistive technology procurement, our findings highlight how decision-making is polycentric rather than clinician-led. By embedding community-driven knowledge into the Assistive-Tech Commons, we formalize these overlooked decision-making influences.

5.1.4 System Perspective on Assistive Technology. Figure 2 synthesizes our findings into an integrative systems perspective. This model captures how *User Agency* emerges not merely from personal capability or assistive devices alone, but through complex negotiations involving *Caregiver Ecology*, *Technology Affordances*, and the broader *Structural Context*. Particularly noteworthy is the reciprocal interaction between caregiving ecologies and technological affordances, demonstrating how caregiving dynamics (e.g., caregiver availability or emotional relationships) can shape the perceived utility of technologies and vice versa.

Additionally, our participants highlighted how structural contexts, especially insurance policies and environmental accessibility, profoundly influenced technology adoption and caregiving arrangements. A critical insight here is the advocacy feedback loop indicated by the arrow from *User Agency* to *Structural Context*, where empowered users drive systemic change, such as through successful advocacy or crowdfunding campaigns, challenging traditional insurance-driven pathways. This systems model builds on socio-technical perspectives such as UTAUT, which recognize the role of social influence in technology adoption, while extending them by incorporating caregiving ecologies and emotional dignity as first-order constraints. It also resonates with Sen’s Capability Approach by emphasizing autonomy as real freedoms to pursue valued ‘beings and doings,’ rather than mere functional use. In doing so, we extend critiques of traditional socio-technical adoption models, which acknowledge social influence but overlook the reciprocal

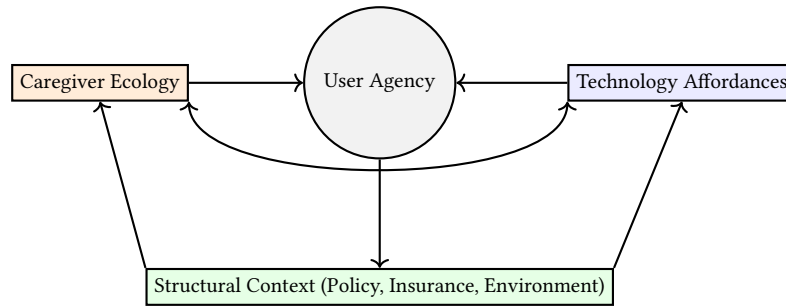


Figure 2: Conceptual systems model showing the bidirectional dependencies among user agency, caregiver ecology, technology affordances, and structural context. The model emphasizes how caregiving ecologies and structural conditions shape the perceived utility of technologies, while empowered users can, in turn, influence policy and systemic change.

role of caregiver ecologies and emotional dignity in shaping the success of technology. By explicitly tying each node of this model to the themes in Section 4, this systems perspective synthesizes our answers to RQ1–RQ3 and identifies where design and policy interventions are most likely to compress ask-points, protect dignity, or redistribute labor.

5.2 Design Implications for a User-Led Design Framework and an Assistive-Tech Commons

Building directly on the patterns surfaced in Section 5.1, we propose two complementary pillars that translate our analysis into a concrete way of working (see Fig. 3). The first pillar is a *Trajectory- and Ask-Point-Centered Co-Design Framework* that situates powered-wheelchair users (PWUs) as co-designers from the outset, ensuring solutions fit the realities of daily life and caregiving ecologies. The second pillar is an *Assistive-Tech Commons*, a user-led infrastructure that consolidates lived knowledge, evidence, and policy scaffolding, enabling promising solutions to enter and persist in everyday practice.

Trajectory- and journey-oriented methods have mapped care transitions and evolving needs in domains such as dementia care and eHealth, revealing opportunities along a patient’s journey [8, 29]. We build on that lineage by making trajectories *operational*, high-frequency ask-points are centered within routines, paired with privacy exposure, and carried forward into procurement and onboarding. Our approach diverges from descriptive journey mapping by tying trajectory metrics to concrete artifacts and by embedding a community backbone that sustains use in the home. For accessibility, we extend previous work on social accessibility, such as dignity, stigma, and fit in social contexts, by treating emotional dignity as a first-order design constraint rather than a downstream consideration [27, 67].

These pillars emerged inductively from three converging signals in our data: (i) high-frequency bottlenecks were concentrated in a handful of daily tasks, (ii) emotional and relational costs strongly shaped acceptance, and (iii) the multi-channel “advice maze” and reimbursement frictions determined acquisition and sustained use. Participants repeatedly emphasized the need for a single place where lived knowledge, costs, and workarounds could be organized by *task*, rather than by brand or technology.

“There needs to be more policy-level work, more subsidies and support, and common ground for sharing knowledge and experiences, because our needs are way beyond the normal needs.” (P8)

This section, therefore, does not introduce an abstract framework in isolation but crystallizes patterns that recurred across participants’ accounts of daily trajectories, emotional externalities, and the advice maze, making the lineage from lived experience to design recommendations explicit.

5.2.1 A User-Led Design Framework. As shown in Fig. 3, the Commons sits at the center of a hub-and-spokes ecology that exchanges knowledge and resources with four surrounding elements: *user agency* (fluid independence in daily life), *caregiver ecology* (family/paid care, rhythms, availability), *technology affordances* (devices, controls, fit), and *structural context* (policy, insurance, environment). The co-design process operationalizes this ecology through a step-wise workflow that integrates discovery, prototyping, evidence-building, and coverage justification, with explicit loops back into community resources. Table 8 makes this linkage concrete by mapping each barrier and enabler from Section 4.4 onto specific principles and steps in the framework, preserving traceability from interview codes to design moves.

Principles.

- *Center lived trajectories (as surfaced in Section 4.1):* Map the cadence of dependence across morning/evening routines and peak moments, not just “can the device do X?” This reveals when help is unavoidable, when delays are most harmful, and where assistance can be safely replaced. It keeps design grounded in how a day actually unfolds in a household, aligning interventions with real pressure points.
- *Reduce ask-points first (targeting the high-frequency activities summarized in Table 4 and the technologies in Table 5):* Focus on the small, frequent requests that accumulate into dependence. Cutting even one or two high-frequency ask-points can free up meaningful minutes, shorten wait times, and prevent interruption cascades. Early, visible wins raise confidence and adherence.
- *Protect emotional dignity (grounded in the affective patterns in Section 4.2 and Table 6):* Treat privacy exposure, surveillance

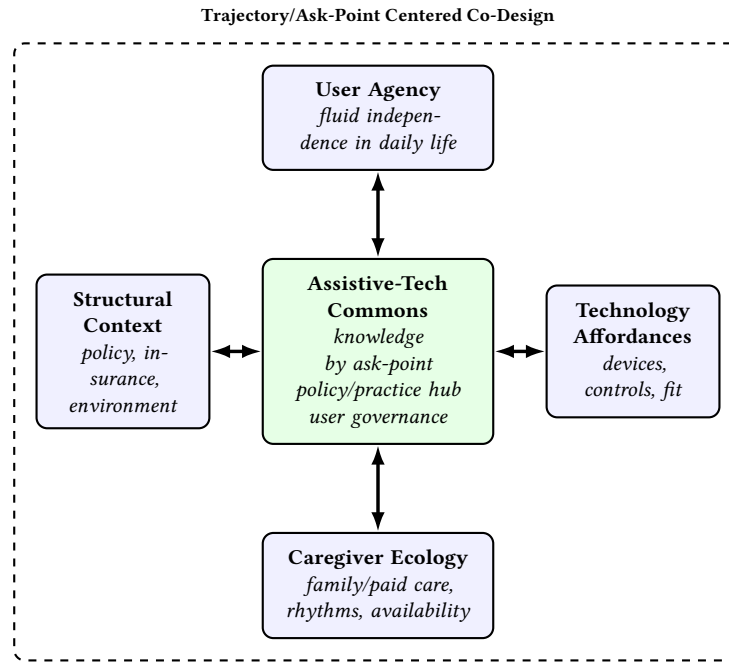


Figure 3: High-level ecosystem view situating the Assistive-Tech Commons as a community-governed hub. Bidirectional spokes represent the dynamic exchanges between user agency, caregiver ecology, technology affordances, and structural context. The model embeds a trajectory- and ask-point-centered co-design cycle, positioning powered-wheelchair users as co-designers and ensuring lived knowledge, policy scaffolding, and emotional dignity are integrated into long-term technology adoption.

feelings, and infantilization risks as primary constraints. Interfaces should preserve user-initiated actions, provide quiet modes, and limit caregiver presence during intimate tasks. Respecting dignity improves acceptance and reduces abandonment when help is nearby.

- *Design with the care ecology (responding to caregiver shortages and rhythms described in Section 4.4):* Solutions must fit caregiver availability, handoffs, and household rhythms, account for who intervenes when devices fail, how shifts overlap, and what training is feasible. Ecological fit keeps devices usable on bad days, not only ideal ones.
- *Be insurance-ready (addressing the insurance denials and opaque pricing discussed in Section 4.4.1):* Generate evidence and documentation while prototyping. Trial data, risk mitigations, and cost comparisons are assembled into insurer-ready packs so adoption does not stall at approval. Building this pathway early reduces downtime and inequity.

Workflow.

- Discovery at peak dependence:* Map morning/evening routines to elicit an Ask-Point Index (API; count of distinct help requests per wake hour by task/urgency) and a Privacy-Gradient Map (task × intimacy with preferred presence: alone/standby/assist). Linking these maps to the **Device Library** surfaces cross-user patterns and quick wins [8].
- Co-design sprints:* PWUs, caregivers, and OTs prototype options that target top ask-points under real constraints like space, reach, mounting, and budget. Artifacts and mounting

patterns move into the **Maker Repository** with clear install notes and safety tiering for reuse [27].

- Evidence & trials:* Short in-home trials capture Δ API, independence minutes gained, caregiver minutes saved, and privacy relief. We log failure modes and backup plans. Outcomes feed the **Policy & Financing** module with routine-level benefits traceable to lived tasks.
- Coverage & policy:* Trial dossiers become insurer-ready packs, such as baseline and post metrics, clinical letters, maintenance plans, and cost comparisons against alternatives. The **Clinician Updater** offers templates and language aligned to coverage criteria to speed submission.
- Onboarding & backup:* Micro-trainings for users and caregivers reduce misuse and downtime. Clear fail-safes define what happens during outages or missed shifts, so independence gains persist under strain. Materials draw from clinician modules and prior Commons cases.
- Feedback:* Results, mounts, and letters flow back to the **Commons** for versioning and peer review. Lived evidence strengthens future approvals and shortens setup time for similar households.

Success Measures. (a) *Autonomy-in-use:* Reduction in ask-points, minutes of self-initiated action gained per day, and caregiver minutes saved. These capture day-shaping gains, not just task success; report medians and interquartile range to reflect skew and outliers.

(b) *Emotional fit*: Privacy relief and dignity scores in sensitive contexts, with notes on presence mode and timing. Scores tie directly to design choices (e.g., user initiation, quiet actions) and predict retention [67].

(c) *System throughput*: Approval rates, days-to-approval, and retention, plus cost per independence hour gained. These indicate whether benefits survive real-world logistics and funding [29].

This framework centers high-frequency ask-points as the unit of analysis, elevates emotional dignity as a primary design constraint, and builds in an explicit pathway to coverage justification and community infrastructure from the outset. Together, these features translate our qualitative findings into a replicable method that addresses RQ3's call for strategies to support sustainable, long-term adoption. We offer this framework as a generative scaffold rather than a fixed protocol. Its specific metrics and modules can be adapted to different clinical and policy contexts, but its core commitment, centered on ask-points, emotional dignity, and coverage pathways, flows directly from our participants' trajectories.

5.2.2 The Assistive-Tech Commons. At the center of Fig. 3, the Commons addresses the fragmentation of knowledge and resources that participants consistently identified as barriers. It organizes lived evidence by *task*, not brand, and ensures that promising solutions persist from discovery to coverage. The desire for such an infrastructure was articulated explicitly by participants, for example, when P52 called for “a single online catalog or database showing all assistive tech and what it can do” (Section 4.4.2).

Online peer communities excel at mutual aid, narration, and stigma-safe exchange but rarely produce structured, trial-based evidence or insurer-ready artifacts [52, 64, 81]. The Commons adds outcome fields, safety tiering, clinician micro-modules, and policy connectors, enabling community knowledge to translate into durable provision. It is not a forum replacement; it is the bridge from story to sustained setup.

- **Device Library**: Entries are indexed by ask-point tags (e.g., reach, door, transfer) with environment constraints, install complexity, typical cost range, maintenance burden, safety tier, failure modes, and user-reported outcomes (e.g., Δ API, minutes gained). This allows households to compare options based on task impact rather than brand or hype.
- **Maker Repository**: Safe, versioned add-ons (mounts, switch layouts, printable parts) carry bill-of-materials, install notes, and a safety tier (A: reversible settings; B: non-load-bearing add-ons; C: load-bearing/transfer-adjacent with professional sign-off). Version history and adverse-event flags protect users while enabling reuse.
- **Policy & Financing**: Typical prices, coverage criteria, common denial reasons, exemplar letters, and crowdfunding connectors. Trial-to-coverage checklists and cost-comparison templates help convert routine-level gains into approval language. Transparency reduces inequities and delays.
- **Clinician Updater**: Brief modules, coding tips, and templates keep OTs current on options and reimbursement practices. Materials align with the workflow's metrics, ensuring that clinical documentation and user evidence speak the same language.

- **User Governance**: A majority-PWUs board sets moderation norms, manages conflicts of interest, and oversees safety reporting and takedowns. Community review signals completeness and reliability without blocking urgent sharing.

The Commons formalizes what wheelchair users already do informally, such as sharing tactics, comparing notes, and exchanging workarounds, while directly linking community expertise with design, policy, and clinical infrastructures. Hence, the framework and the Commons shift the focus of assistive technology design from episodic interventions toward sustained, ecosystem-level practices. Rather than treating wheelchair users as end-points of innovation, they position users, caregivers, and communities as active co-creators whose lived expertise directly shapes procurement, onboarding, and long-term integration. Our framework operationalizes ideals from participatory and justice-oriented approaches by embedding them into trajectory-centered methods and a sustainable Commons infrastructure. This directly addresses the gap that participatory projects often remain episodic or fragmented, failing to translate co-design principles into durable, community-governed systems. This also advances beyond prior interview-based studies with PWUs, which surfaced important insights but remained device- or scenario-bounded. By introducing repeatable practices such as the Ask-Point Index and Commons infrastructure, we translate qualitative insights into actionable, sustainable systems. Where prior work has documented peer exchange, maker cultures, or information portals in isolation, the Assistive-Tech Commons combines these elements into a single, polycentric infrastructure that aligns clinical evidence, user experience, DIY ingenuity, and policy scaffolding around concrete ask-points, offering a systems-level response to the episodic impact of many participatory assistive technology projects. We should always be mindful and respectful when designing such technologies, recognizing that their impact extends far beyond specific conditions. Any of us may need these supports in the near future, as one participant powerfully stated:

“I always say every person is one minute away from becoming disabled themselves... Universal design is just smart, and it's not about disability.” (P46)

5.3 Implications for Research on Assistive Technologies

Our findings surface several critical gaps and opportunities, outlining clear trajectories for advancing HCI research in assistive technology. Building on the lived experiences and systemic challenges identified in this study, we propose five strategic directions:

5.3.1 Longitudinal and Cross-Cultural Studies of Autonomy Dynamics. Our findings highlight that independence is not a binary state measured only by hours of unsupervised activity, but a fluid and continually negotiated experience shaped by technologies, caregiving ecologies, and structural conditions. To advance this perspective, future research should develop methods that capture the temporal oscillations of autonomy, such as longitudinal diaries, sensor-based logging, or experience sampling, to reveal how independence and dependency shift across daily routines, caregiving rhythms, and life stages [12]. Cross-cultural comparisons are equally critical, as

Barrier from findings	Framework element	How it helps
Cost & insurance denial	Coverage justification pack; Commons transparency tools; crowdfunding playbooks	Converts lived tasks into reimbursable claims; reduces opaque pricing
Caregiver scarcity	Design with care ecology; fail-safe protocols	Sustains use when help is unavailable
Training & knowledge gaps	Micro-onboarding; OT upskilling; peer video trials	Reduces abandonment; keeps clinicians current
Design/reliability issues	Home trials; mod repository; adverse-event logging	Captures failures in context; iterates toward fit
Policy & access frictions	Commons policy briefs; template letters; local resource maps	Speeds appeals; connects to transport/access fixes

Table 8: Mapping of barriers identified by participants to specific elements of the proposed Trajectory- and Ask-Point-Centered Co-Design Framework and Assistive-Tech Commons. The table illustrates how framework components address systemic obstacles.

norms around caregiving, systemic inequities, and policy infrastructures shape adoption differently across contexts. Thus, these directions build on our concept of fluid independence by grounding it empirically in both temporal and cultural variability, providing a richer account of autonomy than current static models allow. Combined with our ask-point-centered analysis and design framework (Section 5.2), such methods would enable future work to trace how specific interventions reshape the oscillations of autonomy and dependence across real daily trajectories.

5.3.2 Participatory and Policy-Oriented Co-Design of Adaptive Technologies. Prior CHI and CSCW work has demonstrated the value of involving wheelchair users in participatory design, from exploring “chairables” as interactive platforms [11] to integrating capture probes and co-analysis methods [63] to reimagining crowd navigation [83]. However, these projects have often been bound by specific devices or scenarios, with limited pathways into policy or reimbursement structures determining long-term sustainability. Our findings highlight that participatory design must extend beyond artifact-focused engagements to address systemic barriers such as insurance denials, caregiver shortages, and training gaps. Future research should therefore couple participatory workshops, in-home prototyping, and evaluations with policy collaborations and insurer partnerships. Our trajectory- and ask-point-centered framework offers one concrete template for such work, embedding coverage justification and policy engagement into the design process from the outset. Building on work in design justice and CBPR [15, 17], such efforts can ensure that user-driven insights not only shape prototypes but also inform structural reforms, embedding dignity, privacy, and caregiving dynamics into sustainable AT ecosystems.

5.3.3 Community-Centered and Emotionally Grounded Infrastructures. Prior research has documented the value of peer-to-peer exchange and DIY maker communities in accessibility contexts [28, 39, 51, 58], as well as the importance of social accessibility and relational autonomy [18, 67, 68]. However, this work has often stopped short of formalizing how such community expertise and affective dimensions can be systematically integrated into the infrastructures that support long-term technology adoption. Our findings show that powered-wheelchair users rely heavily on peers, expos,

and DIY solutions and evaluate technologies regarding dignity, privacy, stigma, and emotional labor, which directly shape adoption and abandonment. Future research should therefore explore user-led infrastructures, such as an “Assistive-Tech Commons”, that consolidate peer reviews, DIY modifications, crowdfunding experiences, and policy workarounds, while also embedding evaluative metrics that capture relational and emotional dimensions (e.g., privacy relief, dignity scores, reduction of ask-points). By institutionalizing community knowledge and affective measures, researchers can move beyond functionality-centered paradigms, extending prior works from documenting peer exchange to creating sustainable, emotionally sensitive ecosystems that democratize access and support lasting inclusion.

These trajectories chart a path toward a more inclusive, empowering, and human-centered assistive technology ecosystem. By extending methodological, design, community, cultural, and policy dimensions, they advance theoretical understanding while delivering systemic impact in the lives of wheelchair users.

6 Limitations and Future Work

Our study represents an early phase in a growing research area, and as such, it carries several limitations. Because this was an exploratory, interview-based study conducted in naturalistic settings rather than a controlled trial, our analyses emphasize descriptive patterns in how powered-wheelchair users negotiate autonomy, caregiving, and technology adoption rather than testing causal mechanisms. We employed rigorous qualitative methods, including iterative coding and peer debriefing, but the reliance on self-reported accounts may introduce recall bias or emphasize more salient experiences over routine ones. Besides, our participant pool was composed exclusively of powered-wheelchair users. While this focus enabled a nuanced understanding of the specific intersections between powered mobility, caregiving, and assistive technology, it necessarily excludes the perspectives of manual-wheelchair users and other mobility aid users, whose experiences may present distinct challenges and design opportunities. Future work should broaden its scope to include these groups and incorporate longitudinal and cross-cultural studies, as well as complementary methods

such as ethnography and observation, to capture how autonomy, dignity, and technology use evolve across contexts and over time.

7 Conclusion

Through rich, first-person narratives from 55 powered-wheelchair users across diverse contexts, this study deepens the understanding of how assistive technology intersects meaningfully and often challengingly with everyday life. Our findings demonstrate that powered-wheelchair use cannot be reduced to device functionality alone, but emerges as a fluid and continually negotiated experience shaped by caregiving dynamics, emotional well-being, and structural conditions. While advanced technologies offered transformative possibilities for autonomy and everyday low-cost tools provided accessible forms of agency, both were mediated by caregiver relationships, systemic barriers, and peer expertise. This complementarity underscores that sustainable assistive technology design must attend to performance and usability, as well as dignity, privacy, and emotional labor. Building on these insights, we proposed design implications that include trajectory- and ask-point-centered co-design, operationalizing emotional and relational metrics, and creating an *Assistive-Tech Commons* to consolidate community knowledge and bridge gaps between users, caregivers, clinicians, and policy systems. More broadly, our study raises questions about how autonomy is conceptualized in HCI, how community-driven expertise can be formalized into durable infrastructures, and how affective and relational factors function as first-order determinants of technology adoption. These implications suggest opportunities for researchers, designers, and policymakers to create assistive technologies and ecosystems that are participatory, emotionally sensitive, and structurally sustainable, positioning powered-wheelchair users not merely as recipients of technology but as co-creators of inclusive futures.

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A Semi-Structured Interview Guide

This appendix presents the semi-structured interview guide used in all interviews with powered-wheelchair users. Interviewers adapted the wording, order, and follow-up questions to participants' narratives while ensuring each topic was covered. Consistent with our ethical protocol, interviewers reminded participants that they could skip any question or stop the interview at any time. Throughout the interview, we used open-ended probes such as “Could you tell me more about that experience?”, “What was going through your mind at that moment?”, and “How did that make you feel?” to deepen and clarify participants' accounts.

A.1 Introduction and Rapport Building

- Brief self-introduction by interviewer(s) and study purpose: *We are interested in learning about your experiences with your powered wheelchair, caregiving, and other technologies you use in daily life.*
- Confirm consent, recording, and confidentiality; reiterate that participants can decline to answer any question.
- Warm-up question:
 - Q1. To start, could you tell me a bit about yourself and what your day-to-day life looks like, in whatever way feels most comfortable to you?

A.2 Daily Routines and Ask-Points

- Q2. Can you walk me through a typical morning in your day, from when you wake up until you feel ready to start your day?
 - Probe: As you describe it, can you point out the moments when you usually need to ask someone for help?
- Q3. Thinking about the rest of your day, what kinds of activities do you usually do?
 - Probe: Where during those times do you most often need assistance?

- Q4. What parts of your daily routine feel the most challenging or dependent on others right now?
- Q5. Are there specific tasks that you wish you could do independently but currently need help with? Could you walk me through an example?

A.3 Caregiving Arrangements and Relationships

- Q6. Who helps you with your day-to-day activities, for example, family members, friends, or paid caregivers? How does that support usually work across a typical day?
- Q7. Can you tell me about a time when receiving help felt especially supportive or worked really well for you?
- Q8. Can you share an example of a time when relying on others for help felt emotionally difficult or uncomfortable?
 - Probe: What was challenging about that situation for you?
- Q9. Have you ever had to wait a long time for help when you really needed it?

A.4 Activities of Daily Living and Independence

- Q10. Thinking about activities like getting out of bed, showering, toileting, dressing, preparing meals, and moving around your home, which ones are most challenging for you right now?
- Q11. When you think about feeling ‘independent’ in your daily life, what does that mean to you personally?
 - Probe: Are there particular moments in the day when you feel especially independent or especially dependent?

A.5 Technology Use in Daily Life

- Q12. What kinds of assistive technologies do you currently use in your daily life?
- Q13. What other technologies, like smartphones or low-tech tools, do you use that help you manage your day?
- Q14. Thinking about a specific device you use a lot, can you tell me the story of how you started using it?
 - Probes: Who first suggested it to you?; What made you decide it was worth trying?

A.6 Decision-Making and Advice Networks

- Q15. When you are considering a new device or technology, how do you usually gather information and decide whether it will be a good fit for you?
 - Probes: Do you talk with therapists, family, peers, or people online?; Are there particular sources you trust more than others?
- Q16. Have you used events like Abilities Expos, online forums, or social media to learn about technologies?

A.7 Structural Barriers and System-Level Factors

- Q17. Have there been times when insurance, funding, or other policies made it hard to get, keep, or repair a device you wanted?
 - Probe: How did you handle that situation?
- Q18. Are there situations where you feel the environment around you makes you more dependent on others than you would like to be?

A.8 Closing

- Q19. Was there a recent moment when you thought, 'If only I had *something*, this would be so much easier'?
- Q20. Is there anything about your experiences with independence, caregiving, or technology that we have not talked about but that you think is important for us to understand?
- Thank participant, confirm compensation, and invite them to follow up if they have additional thoughts after the interview.